

Internship Project — Week 1

Project Title: House Price Prediction

Project Summary Report

Objective

The objective of this project was to build a machine learning model capable of predicting house prices based on various property features such as area, number of bedrooms, bathrooms, stories, parking availability, and other amenities. The project also aimed to identify the factors that have the greatest influence on house prices and compare the performance of different regression models.

Dataset and Data Preparation

The Housing Prices dataset was loaded using Pandas and initially explored to understand its structure and contents. The dataset contained 545 records and 13 columns, with the target variable being **price**. The dataset was checked for missing values and duplicate records. No missing values or duplicate rows were found.

Several columns contained categorical values such as *mainroad*, *guestroom*, *basement*, *airconditioning*, and *furnishingstatus*. These categorical variables were converted into numerical form using one-hot encoding so that they could be used effectively by machine learning algorithms. After preprocessing, the dataset contained 14 columns and was ready for model training.

Model Development and Evaluation

The dataset was divided into training and testing sets using an 80:20 split. Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The performance of both models was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score.

Model	MAE	RMSE	R ² Score
Linear Regression	970,043.40	1,324,506.96	0.6529
Random Forest Regressor	1,021,546.04	1,400,565.97	0.6119

The results showed that the Linear Regression model performed better than the Random Forest Regressor on this dataset. The Linear Regression model achieved an R² score of 0.6529, indicating that it explained approximately 65% of the variation in house prices.

Key Findings

The visualizations and correlation analysis revealed that features such as **area**, **bathrooms**, **stories**, **parking**, and **air conditioning** had a noticeable impact on house prices. Among all features, house area appeared to be one of the strongest factors influencing property value. Houses with larger areas and additional amenities generally had higher prices.

One interesting observation was that the simpler Linear Regression model outperformed the more complex Random Forest model. This suggests that the relationship between the selected features and house prices is relatively linear within this dataset.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. After comparing both models, Linear Regression was identified as the better-performing model for this dataset. The findings highlight the importance of property size and essential amenities in determining house prices. Based on the analysis, real estate businesses should focus on emphasizing factors such as area, parking facilities, air conditioning, and number of bathrooms when valuing and marketing residential properties.